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Artificial Intelligence and the Future of Teaching and Learning

Insights and Recommendations

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Artificial Intelligence and the Future of Teaching and Learning

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Table of Contents

Introduction **1**

Rising Interest in AI in Education 1
Three Reasons to Address AI in Education Now 2
Toward Policies for AI in Education 3

Building Ethical, Equitable Policies Together **6**

Guiding Questions 6
Foundation 1: Center People (Parents, Educators, and Students) 6
Foundation 2: Advance Equity 7
Foundation 3: Ensure Safety, Ethics, and Effectiveness 8
Foundation 4: Promote Transparency 9
Overview of Document 10

What is AI? **11**

Perspective: Human-Like Reasoning 12
Perspective: An Algorithm that Pursues a Goal 12
Perspective: Intelligence Augmentation 14
Definition of "Model" 14
Insight: AI Systems Enable New Forms of Interaction 15
Key Recommendation: Human in the Loop AI 16

Learning **18**

Insight: AI Enables Adaptivity in Learning 18
Intelligent Tutoring Systems: An Example of AI Models 19
Important Directions for Expanding AI-Based Adaptivity 20
A Duality: Learning With and About AI 22
A Challenge: Systems Thinking About AI in Education 22
Open Questions About AI for Learning 23
Key Recommendation: Seek AI Models Aligned to a Vision for Learning 24

Teaching **25**

Always Center Educators in Instructional Loops 25
Insight: Using AI to Improve Teaching Jobs 26
Preparing and Supporting Teachers in Planning and Reflecting 29
Designing, Selecting, and Evaluating AI Tools 30
Challenge: Balancing Human and Computer Decision-Making 30
Challenge: Making Teaching Jobs Easier While Avoiding Surveillance 31
Challenge: Responding to Students' Strengths While Protecting Their Privacy 32
Questions Worth Asking About AI for Teaching 34
Key Recommendation: Inspectable, Explainable, Overridable AI 34

Formative Assessment 37
Building on Best Practices 37
Implications for Teaching and Learning 38
Insight: AI Can Enhance Feedback Loops 39
An Example: Automated Essay Scoring 40
Key Opportunities for AI in Formative Assessment 41
Key Recommendation: Harness Assessment Expertise to Reduce Bias 42
Related Questions 43
Research and Development 44
Insight: Research Can Strengthen the Role of Context in AI 44
Attention to the Long Tail of Learner Variability 46
Partnership in Design-Based Research 47
Re-thinking Teacher Professional Development 48
Connecting with Public Policy 49
Key Recommendation: Focus R&D on Addressing Context 50
Ongoing Questions for Researchers 50
Desired National R&D Objectives 51
Recommendations 52
Insight: Aligning AI to Policy Objectives 52
Calling Education Leaders to Action 53
Recommendation #1: Emphasize Humans in the Loop 53
Recommendation #2: Align AI Models to a Shared Vision for Education 54
Recommendation #3: Design Using Modern Learning Principles 56
Recommendation #4: Prioritize Strengthening Trust 57
Recommendation #5: Inform and Involve Educators 57
Recommendation #6: Focus R&D on Addressing Context and Enhancing Trust and Safety 59
Recommendation #7: Develop Education-Specific Guidelines and Guardrails 60
Next Steps 60
Common Acronyms and Abbreviations 62
Acknowledgements 63
References 64

Introduction

The U.S. Department of Education (Department) is committed to supporting the use of technology to improve teaching and learning and to support innovation throughout educational systems. This report addresses the clear need for sharing knowledge and developing policies for "Artificial Intelligence," a rapidly advancing class of foundational capabilities which are increasingly embedded in all types of educational technology systems and are also available to the public. We will consider "educational technology" (edtech) to include both (a) technologies specifically designed for educational use, as well as (b) general technologies that are widely used in educational settings. Recommendations in this report seek to engage teachers, educational leaders, policy makers, researchers, and educational technology innovators and providers as they work together on pressing policy issues that arise as Artificial Intelligence (AI) is used in education.

AI can be defined as "automation based on associations." When computers automate reasoning based on associations in data (or associations deduced from expert knowledge), two shifts fundamental to AI occur and shift computing beyond conventional edtech: (1) from capturing data to *detecting patterns* in data and (2) from providing access to instructional resources to *automating decisions* about instruction and other educational processes. Detecting patterns and automating decisions are leaps in the level of responsibilities that can be delegated to a computer system. The process of developing an AI system may lead to bias in how patterns are detected and unfairness in how decisions are automated. Thus, educational systems must govern their use of AI systems. This report describes opportunities for using AI to improve education, recognizes challenges that will arise, and develops recommendations to guide further policy development.

Rising Interest in AI in Education

Today, many priorities for improvements to teaching and learning are unmet. Educators seek technology-enhanced approaches addressing these priorities that would be safe, effective, and scalable. Naturally, educators wonder if the rapid advances in technology in everyday lives could help. Like all of us, educators use AI-powered services in their everyday lives, such as voice assistants in their homes; tools that can correct grammar, complete sentences, and write essays; and automated trip planning on their phones. Many educators are actively exploring AI tools as they are newly released to the public[1]. Educators see opportunities to use AI-powered capabilities like speech recognition to increase the support available to students with disabilities, multilingual learners, and others who could benefit from greater adaptivity and personalization in digital tools for learning. They are exploring how AI can enable writing or improving lessons, as well as their process for finding, choosing, and adapting material for use in their lessons.

Educators are also aware of new risks. Useful, powerful functionality can also be accompanied with new data privacy and security risks. Educators recognize that AI can automatically produce output that is inappropriate or wrong. They are wary that the associations or automations created by AI may amplify unwanted biases. They have noted new ways in which students may

[1] Walton Family Foundation (March 1, 2023). Teachers and students embrace ChatGPT for education. https://www.waltonfamilyfoundation.org/learning/teachers-and-students-embrace-chatgpt-for-education

represent others' work as their own. They are well-aware of "teachable moments" and pedagogical strategies that a human teacher can address but are undetected or misunderstood by AI models. They worry whether recommendations suggested by an algorithm would be fair. Educators' concerns are manifold. Everyone in education has a responsibility to harness the good to serve educational priorities while also protecting against the dangers that may arise as a result of AI being integrated in edtech.

To develop guidance for edtech, the Department works closely with educational constituents. These constituents include educational leaders—teachers, faculty, support staff, and other educators—researchers; policymakers; advocates and funders; technology developers; community members and organizations; and, above all, learners and their families/caregivers. Recently, through its activities with constituents, the Department noticed a sharp rise in interest and concern about AI. For example, a 2021 field scan found that developers of all kinds of technology systems—for student information, classroom instruction, school logistics, parent-teacher communication, and more—expect to add AI capabilities to their systems. Through a series of four listening sessions conducted in June and August 2022 and attended by more than 700 attendees, it became clear that constituents believe that action is required now in order to get ahead of the expected increase of AI in education technology—and they want to roll up their sleeves and start working together. In late 2022 and early 2023, the public became aware of new generative AI chatbots and began to explore how AI could be used to write essays, create lesson plans, produce images, create personalized assignments for students, and more. From public expression in social media, at conferences, and in news media, the Department learned more about risks and benefits of AI-enabled chatbots. And yet this report will not focus on a specific AI tool, service, or announcement, because AI-enabled systems evolve rapidly. Finally, the Department engaged the educational policy expertise available internally and in its relationships with AI policy experts to shape the findings and recommendations in this report.

Three Reasons to Address AI in Education Now

> *"I strongly believe in the need for stakeholders to understand the cyclical effects of AI and education. By understanding how different activities accrue, we have the ability to support virtuous cycles. Otherwise, we will likely allow vicious cycles to perpetuate."*
> *—Lydia Liu*

During the listening sessions, constituents articulated three reasons to address AI now:

First, AI may enable achieving educational priorities in better ways, at scale, and with lower costs. Addressing varied unfinished learning of students due to the pandemic is a policy priority, and AI may improve the adaptivity of learning resources to students' strengths and needs. Improving teaching jobs is a priority, and via automated assistants or other tools, AI may provide teachers greater support. AI may also enable teachers to extend the support they offer to individual students when they run out of time. Developing resources that are responsive to the knowledge and experiences students bring to their learning—their community and cultural assets—is a priority, and AI may enable greater customizability of curricular resources to meet local needs.

As seen in voice assistants, mapping tools, shopping recommendations, essay-writing capabilities, and other familiar applications, AI may enhance educational services.

Second, urgency and importance arise through awareness of system-level risks and anxiety about potential future risks. For example, students may become subject to greater surveillance. Some teachers worry that they may be replaced—to the contrary, the Department firmly rejects the idea that AI could replace teachers. Examples of discrimination from algorithmic bias are on the public's mind, such as a voice recognition system that doesn't work as well with regional dialects, or an exam monitoring system that may unfairly identify some groups of students for disciplinary action. Some uses of AI may be infrastructural and invisible, which creates concerns about transparency and trust. AI often arrives in new applications with the aura of magic, but educators and procurement policies require that edtech show efficacy. AI may provide information that appears authentic, but actually is inaccurate or lacking a basis in reality. Of the highest importance, AI brings new risks in addition to the well-known data privacy and data security risks, such as the risk of scaling pattern detectors and automations that result in "algorithmic discrimination" (e.g., systematic unfairness in the learning opportunities or resources recommended to some populations of students).

Third, urgency arises because of the scale of possible unintended or unexpected consequences. When AI enables instructional decisions to be automated at scale, educators may discover unwanted consequences. In a simple example, if AI adapts by speeding curricular pace for some students and by slowing the pace for other students (based on incomplete data, poor theories, or biased assumptions about learning), achievement gaps could widen. In some cases, the quality of available data may produce unexpected results. For example, an AI-enabled teacher hiring system might be assumed to be more objective than human-based résumé scoring. Yet, if the AI system relies on poor quality historical data, it might de-prioritize candidates who could bring both diversity and talent to a school's teaching workforce.

In summary, it is imperative to address AI in education *now* to realize key opportunities, prevent and mitigate emergent risks, and tackle unintended consequences.

Toward Policies for AI in Education

The 2023 AI Index Report from the Stanford Institute for Human-Centered AI has documented notable acceleration of investment in AI as well as an increase of research on ethics, including issues of fairness and transparency.[2] Of course, research on topics like ethics is increasing because problems are observed. Ethical problems will occur in education, too.[3] The report found a striking interest in 25 countries in the number of legislative proposals that specifically include AI. In the United States, multiple executive orders are focused on ensuring AI is trustworthy and equitable, and the White House Office of Science and Technology Policy has introduced a

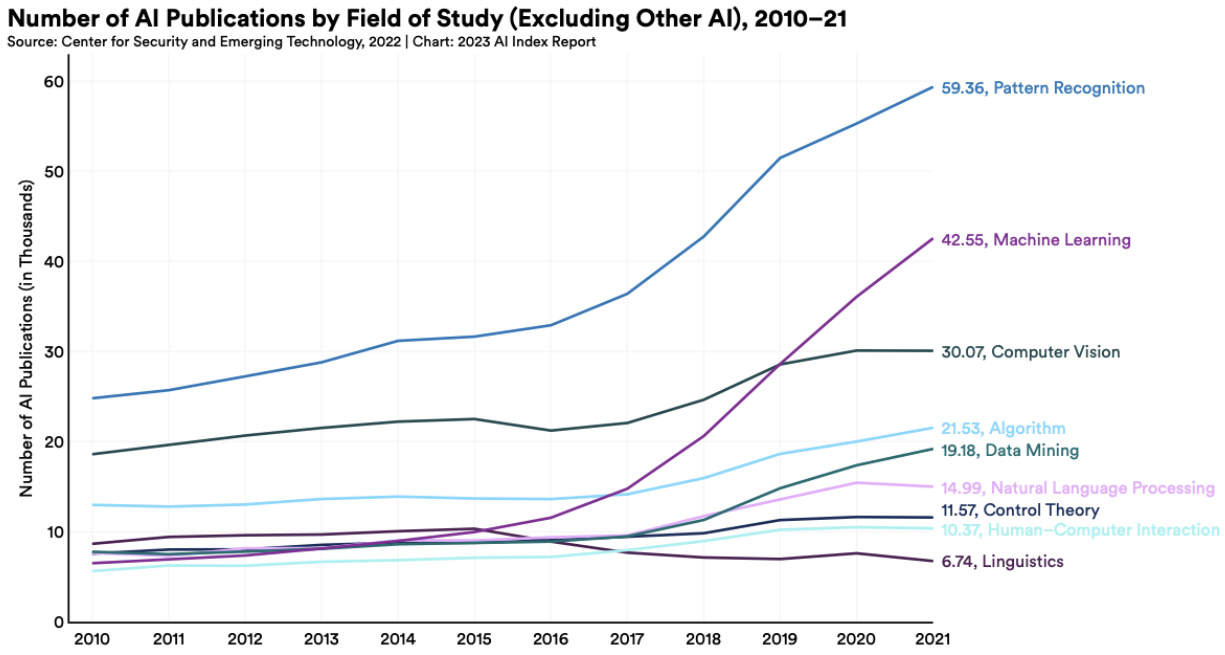
[2] Maslej, N., Fattorini, L., Brynjolfsson E., Etchemendy, J., Ligett, K., Lyons, T., Manyika, J., Ngo, H., Niebles, J.C., Parli, V., Shoham, Y., Wald, R., Clark, J. and Perrault, R., (2023). *The AI index 2023 annual report.* Stanford University: AI Index Steering Committee, Institute for Human-Centered AI.

[3] Holmes, W. & Porayska-Pomsta, K. (Eds.) (2022). *The ethics of artificial intelligence in education.* Routledge. ISBN 978-0367349721

Blueprint for an AI Bill of Rights (Blueprint)[4] that provides principles and practices that help achieve this goal. These initiatives, along with other AI-related policy activities occurring in both the executive and legislative branches, will guide the use of AI throughout all sectors of society. In Europe, the European Commission recently released *Ethical guidelines on the use of artificial intelligence (AI) and data in teaching and learning for educators*.[5]

AI is moving fast and heralding societal changes that require a national policy response. In addition to broad policies for all sectors of society, education-specific policies are needed to address new opportunities and challenges within existing frameworks that take into consideration federal student privacy laws (such as the Family Educational Rights and Privacy Act, or FERPA), as well as similar state related laws. AI also makes recommendations and takes actions automatically in support of student learning, and thus educators will need to consider how such recommendations and actions can comply with laws such as the Individuals with Disabilities Education Act (IDEA). We discuss specific policies in the concluding section.

Figure 1: Research about AI is growing rapidly. Other indicators, such as dollars invested and number of people employed, show similar trends.



AI is advancing exponentially (see Figure 1), with powerful new AI features for generating images and text becoming available to the public, and leading to changes in how people create text and

[4] White House Office of Science and Technology Policy (October 2022), *Blueprint for an AI bill of rights: Making automated systems work for the American people.* The White House Office of Science and Technology Policy. https://www.whitehouse.gov/ostp/ai-bill-of-rights/

[5] European Commission, Directorate-General for Education, Youth, Sport and Culture. (2022). *Ethical guidelines on the use of artificial intelligence (AI) and data in teaching and learning for educators*, Publications Office of the European Union. **https://data.europa.eu/doi/10.2766/153756**

images[6]. The advances in AI are not only happening in research labs but also are making news in mainstream media and in educational-specific publications.

Researchers have articulated a range of concepts and frameworks for ethical AI[7], as well as for related concepts such as equitable, responsible, and human-centered AI. Listening session participants called for building on these concepts and frameworks but also recognized the need to do more; participants noted a pressing need for guardrails and guidelines that make educational use of AI advances safe, especially given this accelerating pace of incorporation of AI into mainstream technologies. As policy development takes time, policy makers and educational constituents together need to start now to specify the requirements, disclosures, regulations, and other structures that can shape a positive and safe future for all constituents—especially students and teachers.

Policies are urgently needed to implement the following:

1. leverage automation to advance learning outcomes while protecting human decision making and judgment;

2. interrogate the underlying data quality in AI models to ensure fair and unbiased pattern recognition and decision making in educational applications, based on accurate information appropriate to the pedagogical situation;

3. enable examination of how particular AI technologies, as part of larger edtech or educational systems, may increase or undermine equity for students; and

4. take steps to safeguard and advance equity, including providing for human checks and balances and limiting any AI systems and tools that undermine equity.

[6] Sharples, M. & Pérez y Pérez, R. (2022). *Story machines: How computers have become creative writers*. Routledge. ISBN 9780367751951

[7] Akgun, S., Greenhow, C. (2022). Artificial intelligence in education: Addressing ethical challenges in K-12 settings. *AI Ethics,* 2, 431–440. https://doi.org/10.1007/s43681-021-00096-7

Building Ethical, Equitable Policies Together

In this report, we aim to build on the listening sessions the Department hosted to engage and inform all constituents involved in making educational decisions so they can prepare for and make better decisions about the role of AI in teaching and learning. AI is a complex and broad topic, and we are not able to cover everything nor resolve issues that still require more constituent engagement. This report is intended to be a starting point.

The opportunities and issues of AI in education are equally important in K-12, higher education, and workforce learning. Due to scope limitations, the examples in this report will focus on K-12 education. The implications are similar at all levels of education, and the Department intends further activities in 2023 to engage constituents beyond K-12 schools.

Guiding Questions

Understanding that AI increases automation and allows machines to do some tasks that only people did in the past leads us to a pair of bold, overarching questions:

1. What is our collective vision of a desirable and achievable educational system that leverages automation to advance learning while protecting and centering human agency?
2. How and on what timeline will we be ready with necessary guidelines and guardrails, as well as convincing evidence of positive impacts, so that constituents can ethically and equitably implement this vision widely?

In the Learning, Teaching, and Assessment sections of this report, we elaborate on elements of an educational vision grounded in what today's learners, teachers, and educational systems need, and we describe key insights and next steps required. Below, we articulate four key foundations for framing these themes. These foundations arise from what we know about the effective use of educational technology to improve opportunity, equity, and outcomes for students and also relate to the new Blueprint.

Foundation 1: Center People (Parents, Educators, and Students)

Education-focused AI policies at the federal, state, and district levels will be needed to guide and empower local and individual decisions about which technologies to adopt and use in schools and classrooms. Consider what is happening in everyday lives. Many of us use AI-enabled products because they are often better and more convenient. For example, few people want to use paper maps anymore; people find that technology helps us plan the best route to a destination more efficiently and conveniently. And yet, people often do not realize how much privacy they are giving up when they accept AI-enabled systems into their lives. AI will bring privacy and other risks that are hard to address only via individual decision making; additional protections will be needed.

There should be clear limits on the ability to collect, use, transfer, and maintain our personal data, including limits on targeted advertising. These limits should put the burden on platforms to minimize how much information they collect, rather than burdening Americans with reading fine print.[8]

As protections are developed, we recommend that policies center people, not machines. To this end, a first recommendation in this document (in the next section) is an emphasis on **AI with humans in the loop**. Teachers, learners, and others need to retain their agency to decide what patterns mean and to choose courses of action. The idea of humans in the loop builds on the concept of "Human Alternatives, Consideration, and Fallback" in the Blueprint and ethical concepts used more broadly in evaluating AI, such as preserving human dignity. A top policy priority must be establishing human in the loop as a requirement in educational applications, despite contrary pressures to use AI as an alternative to human decision making. Policies should not hinder innovation and improvement, nor should they be burdensome to implement. Society needs an education-focused AI policy that protects civil rights and promotes democratic values in the building, deployment, and governance of automated systems to be used across the many decentralized levels of the American educational system.

Foundation 2: Advance Equity

"AI brings educational technology to an inflection point. We can either increase disparities or shrink them, depending on what we do now."
—Dr. Russell Shilling

A recent Executive Order[9] issued by President Biden sought to strengthen the connection among racial equity, education and AI, stating that "members of underserved communities—many of whom have endured generations of discrimination and disinvestment—still confront significant barriers to realizing the full promise of our great Nation, and the Federal Government has a responsibility to remove these barriers" and that the Federal Government shall both "pursue educational equity so that our Nation's schools put every student on a path to success" and also "root out bias in the design and use of new technologies, such as artificial intelligence." A specific vision of equity, such as described in the Department's recent report, *Advancing Digital Equity for All*[10] is essential to policy discussion about AI in education. This report defines digital equity as

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[8] The White House (September 8, 2022). Readout of White House listening session on tech platform accountability. https://www.whitehouse.gov/briefing-room/statements-releases/2022/09/08/readout-of-white-house-listening-session-on-tech-platform-accountability/

[9] The White House (February 17, 2023). Executive order on further advancing racial equity and support for underserved communities through the federal government. https://www.whitehouse.gov/briefing-room/presidential-actions/2023/02/16/executive-order-on-further-advancing-racial-equity

[10] U.S. Department of Education, Office of Educational Technology (2022). Advancing digital equity for all: Community-based recommendations for developing effective digital equity plans to close the digital divide and enable technology-empowered learning. US Department of Education.

"the condition in which individuals and communities have the information technology capacity that is needed for full participation in the society and economy of the United States."

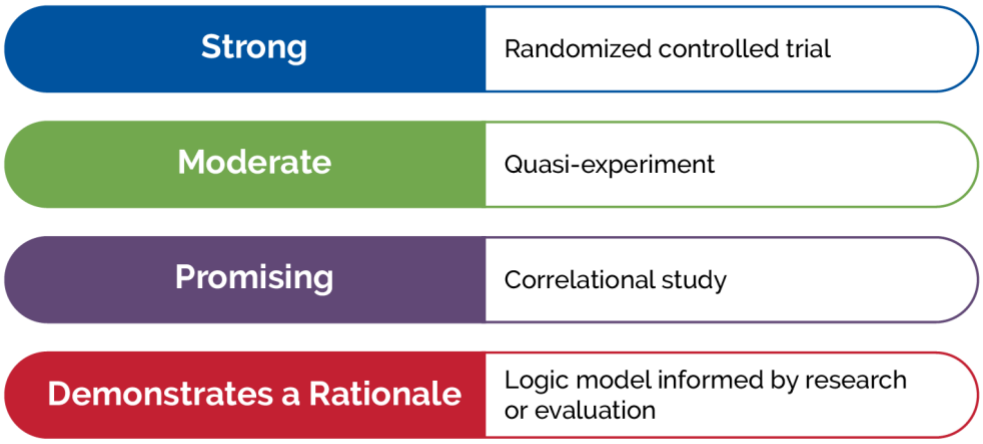
Issues related to racial equity and unfair bias were at the heart of every listening session we held. In particular, we heard a conversation that was increasingly attuned to issues of data quality and the consequences of using poor or inappropriate data in AI systems for education. Datasets are used to develop AI, and when they are non-representative or contain undesired associations or patterns, resulting AI models may act unfairly in how they detect patterns or automate decisions. Systematic, unwanted unfairness in how a computer detects patterns or automates decisions is called "algorithmic bias." Algorithmic bias could diminish equity at scale with unintended discrimination. As this document discussed in the *Formative Assessment* section, this is not a new conversation. For decades, constituents have rightly probed whether assessments are unbiased and fair. Just as with assessments, whether an AI model exhibits algorithmic bias or is judged to be fair and trustworthy is critical as local school leaders make adoption decisions about using AI to achieve their equity goals.

We highlight the concept of "algorithmic discrimination" in the Blueprint. Bias is intrinsic to how AI algorithms are developed using historical data, and it can be difficult to anticipate all impacts of biased data and algorithms during system design. The Department holds that biases in AI algorithms must be addressed when they introduce or sustain unjust discriminatory practices in education. For example, in postsecondary education, algorithms that make enrollment decisions, identify students for early intervention, or flag possible student cheating on exams must be interrogated for evidence of unfair discriminatory bias—and not only when systems are designed, but also later, as systems become widely used.

Foundation 3: Ensure Safety, Ethics, and Effectiveness

A central safety argument in the Department's policies is the need for data privacy and security in the systems used by teachers, students, and others in educational institutions. The development and deployment of AI requires access to detailed data. This data goes beyond conventional student records (roster and gradebook information) to detailed information about what students do as they learn with technology and what teachers do as they use technology to teach. AI's dependence on data requires renewed and strengthened attention to data privacy, security, and governance (as also indicated in the Blueprint). As AI models are not generally developed in consideration of educational usage or student privacy, the educational application of these models may not be aligned with the educational institution's efforts to comply with federal student privacy laws, such as FERPA, or state privacy laws.

Figure 2: The Elementary and Secondary Education Act defines four levels of evidence.



Further, educational leaders are committed to basing their decisions about the adoption of educational technology on evidence of effectiveness—a central foundation of the Department's policy. For example, the requirement to base decisions on evidence also arises in the Elementary and Secondary Education Act (ESEA), as amended, which introduced four tiers of evidence (see Figure 2). Our nation's research agencies, including the Institute of Education Sciences, are essential to producing the needed evidence. The Blueprint calls for evidence of effectiveness, but the education sector is ahead of that game: we need to insist that AI-enhanced edtech rises to meet ESEA standards as well.

Foundation 4: Promote Transparency

The central role of complex AI models in a technology's detection of patterns and implementation of automation is an important way in which AI-enabled applications, products, and services will be different from conventional edtech. The Blueprint introduces the need for transparency about AI models in terms of disclosure ("notice") and explanation. In education, decision makers will need more than notice—they will need to understand how AI models work in a range of general educational use cases, so they can better anticipate limitations, problems, and risks.

AI models in edtech will be approximations of reality and, thus, constituents can always ask these questions: **How precise are the AI models? Do they accurately capture what is most important? How well do the recommendations made by an AI model fit educational goals? What are the broader implications of using AI models at scale in educational processes?**

Building on what was heard from constituents, the sections of this report develop the theme of evaluating the quality of AI systems and tools using multiple dimensions as follows:

- **About AI**: AI systems and tools must respect data privacy and security. Humans must be in the loop.
- **Learning**: AI systems and tools must align to our collective vision for high-quality learning, including equity.
- **Teaching**: AI systems and tools must be inspectable, explainable, and provide human alternatives to AI-based suggestions; educators will need support to exercise professional judgment and override AI models, when necessary.

- **Formative Assessment**: AI systems and tools must minimize bias, promote fairness, and avoid additional testing time and burden for students and teachers.
- **Research and Development**: AI systems and tools must account for the context of teaching and learning and must work well in educational practice, given variability in students, teachers, and settings.
- **Recommendations**: Use of AI systems and tools must be safe and effective for students. They must include algorithmic discrimination protections, protect data privacy, provide notice and explanation, and provide a recourse to humans when problems arise. The people most affected by the use of AI in education must be part of the development of the AI model, system, or tool, even if this slows the pace of adoption.

We return to the idea that these considerations fit together in a comprehensive perspective on the quality of AI models in the *Recommendations* section.

Overview of Document

We begin in the next section by elaborating a definition of AI, followed by addressing learning, teaching, assessment, and research and development. Organizing key insights by these topics keeps us focused on exploring implications for improving educational opportunity and outcomes for students throughout the report.

Within these topics, three important themes are explored:

1. **Opportunities and Risks.** Policies should focus on the most valuable educational advances while mitigating risks.

2. **Trust and Trustworthiness.** Trust and safeguarding are particularly important in education because we have an obligation to keep students out of harm's way and safeguard their learning experiences.

3. **Quality of AI Models.** The process of developing and then applying a model is at the heart of any AI system. Policies need to support evaluation of the qualities of AI models and their alignment to goals for teaching and learning during the processes of educational adoption and use.

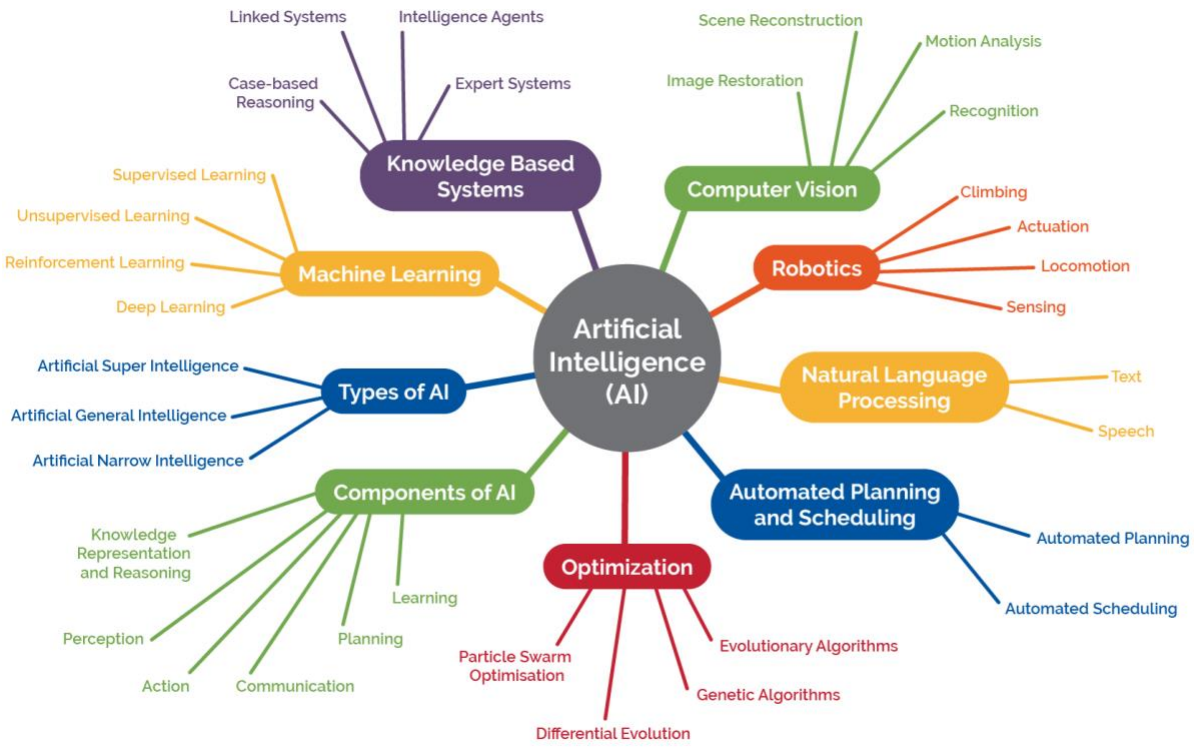
"AI in education can only grow at the speed of trust."
—Dr. Dale Allen

What is AI?

Our preliminary definition of AI as automation based on associations requires elaboration. Below we address three additional perspectives on what constitutes AI. Educators will find these different perspectives arise in the marketing of AI functionality and are important to understand when evaluating edtech systems that incorporate AI. One useful glossary of AI for Education terms is the CIRCLS Glossary of Artificial Intelligence Terms for Educators.[11]

AI is not one thing but an umbrella term for a growing set of modeling capabilities, as visualized in Figure 3.

Figure 3: Components, types, and subfields of AI based on Regona et al (2022).[12]



[11] Search for "AI Glossary Educators" to find other useful definitions.

[12] Regona, Massimo & Yigitcanlar, Tan & Xia, Bo & Li, R.Y.M. (2022). Opportunities and adoption challenges of AI in the construction industry: A PRISMA review. *Journal of Open Innovation Technology Market and Complexity, 8(45).* https://doi.org/10.3390/joitmc8010045